

Module: 13- Networking with Windows Server

1. Discuss the role of Windows Firewall in Windows Server and how to configure it.

Ans : Windows Defender Firewall helps protect Windows Server by monitoring and controlling incoming and outgoing network traffic based on security rules.

- Open Server Manager.
- Go to Tools → Windows Defender Firewall with Advanced Security.
- Select:
 - Inbound Rules
 - Outbound Rules
- Click New Rule.
- Choose the rule type:
 - Program
 - Port
 - Predefined
 - Custom
- Specify the port number or application.
- Choose the action:
 - Allow the connection
 - Block the connection
- Select the network profile (Domain, Private, Public).
- Enter a rule name and click Finish.
- Enable, disable, modify, or delete rules as needed.
- Test connectivity to verify firewall configuration.

2. What is Network Address Translation (NAT) in Windows Server, and how do you configure it?

Ans : NAT helps conserve IP addresses and improves network security by hiding internal IP addresses.

- Open Server Manager.
- Click Add Roles and Features.
- Install the Remote Access role.
- Under Remote Access, select Routing.
- Complete the installation and restart the server if required.
- Open Routing and Remote Access from Tools.
- Right-click the server name and select Configure and Enable Routing and Remote Access.
- Choose Network Address Translation (NAT).
- Select the network interface connected to the Internet.
- Enable Public interface connected to the Internet.
- Select the private network interface.
- Finish the wizard and start the service.
- Verify Internet connectivity from client computers.

3. Explain the concept of Dynamic Host Configuration Protocol (DHCP) and how to configure it in Windows Server 2016.

Ans : Dynamic Host Configuration Protocol is a network service that automatically assigns IP addresses and other network settings to client computers. It reduces manual configuration and simplifies network management.

- Open Server Manager.
- Click Add Roles and Features.
- Select DHCP Server role and install it.
- Complete the DHCP Post-Install Configuration.
- Open DHCP from Tools.
- Expand the server name.

- Right-click IPv4 and select New Scope.
- Enter the scope name and description.
- Specify the IP address range.
- Configure subnet mask.
- Add exclusion ranges if needed.
- Set lease duration.
- Configure router (default gateway) address.
- Configure DNS server and domain name.
- Activate the scope.
- Verify that client computers receive IP addresses automatically.

4. Describe the process of configuring DNS (Domain Name System) in Windows Server.

Ans : Domain Name System is a service used to translate domain names into IP addresses. In Windows Server, DNS is essential for network communication and Active Directory operations.

5. What is Server Manager, and how do you use it to manage servers in Windows Server?

Ans : Server Manager is a management tool in Windows Server used to configure, monitor, and manage local and remote servers from a centralized interface.

It helps administrators manage server roles, features, services, storage, networking, and system status efficiently.

6. Discuss the role of Remote Desktop Services (RDS) in Windows Server 2016 or 2019 and how to configure it.

Ans : Remote Desktop Services is a Windows Server role that allows users to remotely access desktops, applications, and server resources over a network.

- Open Server Manager.
- Select Remote Desktop Services Installation.
- Choose deployment type (Quick Start or Standard Deployment).
- Install required RDS role services
- Complete the installation and restart the server if required.
- Open Remote Desktop Licensing Manager.
- Activate the RDS license server.
- Enable Remote Desktop from Server Manager → Local Server.
- Add authorized users or groups for remote access.
- Configure Windows Defender Firewall to allow Remote Desktop connections.
- Ensure port 3389 is open.
- Enter the server IP address or hostname.